

**Paper Pattern: There will be 52 questions**

**MCQs: 40      2 Marks: 4      3 Marks: 4      5 Marks: 4**

**Time Duration: 120 minutes.**

**Syllabus: Lesson 21-42(Topics 104-209)**

# Objectives

Q. In syllable timed languages, all syllables tend to have an **equal** time value (for example, their length or duration) and the rhythm of the language is said to be syllable-timed

Q. TESOL stand for **Teaching English to Speakers of Other Languages**

Q. ELTR is a project of: **higher education commission**

Q. stress time language have **(stress)** as their dominating rhythmic features.

Q. In spectrograms, time runs from left to right, the frequency of the components is shown on the **vertical** scale.

Q. acoustics is the study of the physics of the **speech signal**

Q. One should observe gap in pattern (with burst for **voiceless** and sharp formant beginning for voiced stops.)

Q. Which of the following vowel is uttered by rounding lips? **/u/**

# Subjective

**Q. spectrogram help to know about bilabial sound.....2 [The most favourite of VU; V.V.IMP for 2 marks questions. They are asking about stops or bilabial or approximants.]**

**Voiced** - vertical striations corresponding to the vibrations of the vocal folds

**Bilabial** - locus of both second and third formants comparatively low

**Alveolar** - locus of second formant about 1700–1800 Hz.

**Velar** - usually high locus of the second formant

**Retroflex** - general lowering of the third and fourth formants

**Stop** - gap in pattern (with burst for voiceless and sharp formant beginning for voiced stops)

**Fricative** - random noise pattern in higher frequency regions

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**Nasal** - formant structure similar to that of vowels (with formants at 250, 2500, and 3250)

**Lateral** - formant structure similar to that of vowels (with formants at 250, 1200, and 2400)

**Approximant** - formant structure similar to that in vowels, usually changing

**Q. Memory for Speech** [Another favourite of VU; V.V.IMP for 2 marks questions. They ask it for speaking style, sound change or any one of them.]

The role of the memory for speech under the exemplar theory suggests that many instances of each word are stored in memory and their phonetic variability is memorized rather than computed. The main postulates of the concepts are given here:

- ✦ **Language universal features:** Broad phonetic classes (e.g., aspirated vs. unaspirated) derive from physiological constraints on speaking or hearing, but their detailed phonetic definitions are arbitrary—a matter of community norms.
- ✦ **Speaking styles:** No one style is basic (from which others are derived), because all are stored in memory. Bilingual speakers store two systems.
- ✦ **Generalization and productivity:** Exemplar theory says that generalization is also possible within productivity. Interestingly, productivity—the hallmark of linguistic knowledge in the phonetic implementation approach—is the least developed aspect of the exemplar theory.
- ✦ **Sound change:** Sound change is phonetically gradual and operates across the whole lexicon. It is a gradual shift as new instances keep on adding.

**Q. Cardinal vowels contrast with primary and secondary**

The primary set includes eight vowels in total (from 1 to 8); the front unrounded vowels [i, e, ε, a], the back unrounded vowel [ɑ] and the rounded back vowels [ɔ, o, u].

A set of secondary cardinal vowels (as a precise set of references) was introduced by the same British phonetician Daniel Jones (1881-1967). Secondary cardinal vowels are easy to understand in connection with the primary cardinal vowel system. The main difference between primary and secondary cardinal vowels is related to lip-rounding as in some languages the feature of lip-rounding is possible for front vowels.

**Q. Explaining the balance between phonetic forces.**

While going for producing sounds with maximum ease of articulation, only similar sounds are affected. The focus of the speakers is always on maintaining a sufficient perceptual distance between the sounds that occur in a contrasting set (e.g., vowels in stressed-monosyllabic words beat, bit, bet, and bat). This principle of perceptual separation does not usually result into one sound affecting an adjacent sound (as explained in the principle of maximum ease of articulation). Instead, perceptual separation affects the set of sounds that potentially can occur at a given position in a word, such as in the position that must be occupied by a vowel in a stressed monosyllable as in words beat, bit, bet, bat so that the perceptual separation is maximized. The principle of ‘maximum perceptual separation’ also accounts for some of the differences between languages. All these examples illustrate how languages maintain a balance between the requirements of the speaker and those of the listener. On the one hand, there is the pressure to make changes that would result in easier articulations from a speaker’s point of view and, then, from the listener’s point of view that there should be sufficient perceptual contrast between sounds that affect the meaning of an utterance.

**Q. IPA**

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While discussing the key elements of linguistic phonetic description, we need to consider the International Phonetic Alphabet (abbreviated as IPA). IPA is the set of symbols and diacritics that have been officially approved by IPA. The association publishes a chart comprising of a number of separate charts. At the top inside the front cover, you will find the main consonant chart. Below it is a table showing the symbols for nonpulmonic consonants, and below that is the vowel chart. Inside the back cover is a list of diacritics and other symbols, and a set of symbols for suprasegmental features (events) such as tone, intonation, stress, and length. Remember that the IPA chart does not try to cover all possible types of phonetic descriptions (e.g., all the individual strategies for realizing linguistic phonological contrasts, or gradations in the degree of co-articulation between adjacent segments, etc.). Instead, it is limited to those possible sounds that can have linguistic significance in that they can change the meaning of a word in some languages. So the description of IPA is based on the linguistic phonetics of the community.

## Q. fundamental distinction consonant and vowel sound?

The fundamental distinction between consonant and vowel sounds is that vowels make the least obstruction to the flow of air. In addition to this, vowels are almost always found at the center of a syllable, and it is very rare to find any sound, other than a vowel which can stand alone as a whole syllable. Phonetically, each vowel has a number of features (properties) that distinguish it from other vowels. These include; firstly, the shape of the lips (lip-rounding), rounded (for sounds like /u:/ vowel), neutral (as for ə - schwa sound) or spread (as in /i:/ sound in word like sea or – when photographers traditionally ask you to say “cheese” /tʃi:z/ in order to make you look smiling. Secondly, part of the tongue - the front, the middle or the back of the tongue may be raised, giving different vowel qualities: compare /æ/ vowel (as in word ‘cat’) as a front vowel, with the /ɑ:/ vowel (as in ‘cart’) which is a back vowel. Thirdly, the tongue (and the lower jaw) may be raised ‘close’ to the roof of the mouth (for close vowels. e.g. /i:/ or /u:/), or the tongue may be left ‘low’ in the mouth with the jaw comparatively ‘open’ (as for open vowels e.g., /a:/ and /æ/. In British phonetics, terms such as ‘close’ and ‘open’ are used for vowels, whereas in American phonetics ‘high’ and ‘low’ are used for vowel description. So, generally, these three aspects are described in the case of vowels; lip-rounding, the part of the tongue and the height of the tongue. In addition to these three features, some other characteristics of vowels are also used in various languages of the world (e.g., nasality – whether a vowel is nasal or not).

## Q. Phonotactics

The study of the phonemes and their order found in the syllables (the study of sound sequences) of a language is called the phonotactics. It has often been found that languages do not allow all phonemes to appear in any order (e.g., a native speaker of English can figure out fairly easily that the sequence of phonemes /streŋθs/ makes an English word (‘strengths’) and that the sequence /bleidg/ would be acceptable as an English word ‘blage’, although that word does not happen to exist, but the sequence /lvm/ could not possibly be the part of an English word). Phonotactic analyses of English come up with some interesting findings. For example, why should ‘bump’, ‘lump’, ‘hump’, ‘rump’, ‘mump(s)’, ‘clump’ and others all be associated with large blunt shapes? Why should there be a whole family of words ending with a plosive and a syllabic /l/ all having meanings to do with clumsy, awkward or difficult action (e.g., ‘muddle’, ‘fumble’, ‘straddle’, ‘cuddle’, ‘fiddle’, ‘buckle’, ‘struggle’, ‘wriggle’)? Why can’t English syllables begin with /pw/, /bw/, /tl/, /dl/ when /pl/, /bl/, /tw/, /dw/ are acceptable? All such discussion is called the phonotactics of the language.

## Q. pitch in tonal language

In tonal languages, pitch is used as an essential component of the pronunciation of a word and a change of pitch may cause a change in meaning. In most languages (whether or not they are tone languages) pitch plays a central role in intonation. In very simple words, pitch is the variation in the vibration of vocal folds.

## Q. Nasal analysis using Paraat

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Formants for nasal sounds are also important for acoustic analysis. Measure the first three (F1, F2 and F3) formants of nasals from the file (use the already learnt way of measuring formants). Remember that nasals have very distinctive waveforms (different than that of vowels) as they have distinctive forms of anti-formants (bands of frequencies damped) and formant transition. When you are done with the measurement, try to answer the following questions:

1. Are there any systematic patterns across nasals?
2. Is there one formant with a similar frequency for all places of articulation?
3. Is there one formant that has much higher amplitude than the others across nasals?
4. Do you see any overall differences between the nasals on the one hand and [a] on the other?

### **Q. nasal formants**

Formants for nasal sounds are also important for acoustic analysis. Measure the first three (F1, F2 and F3) formants of nasals from the file (use the already learnt way of measuring formants). Remember that nasals have very distinctive waveforms (different than that of vowels) as they have distinctive forms of anti-formants (bands of frequencies damped) and formant transition.

### **Q. Speaker styles?**

A complete range of a speaker's vowel qualities may be considered as representative of the speaker's personal features which, in turn, may be compared with the formant frequency of each vowel (with the total range of that formant in that speaker's voice). But this is true that the phoneticians are still working with comparing the acoustic data of one individual with the other and improve further the system of speech recognition. Experts of applied phonetics and computer speech technology are trying to understand the complexity of speech – synthesis systems and improve it.

### **Q. What steps will you follow to calculate Vot on praat**

To calculate the VOT. Record /apa/, /aba/, /ata/, /ada/, /apha/ and /atha/. Zoom in through your stop sounds so that you can analyze the patterns of the stop sounds and find the difference among the three types of VOT (negative, zero and positive). Measure the VOT of each stop and compare voiced/voiceless counterparts (p/b, t/d, k/g). Similarly, zoom in so that you can clearly see the stop closure followed by the beginning of the vowel. You can measure the time between the end of the stop closure (the beginning of the release burst) and the onset of voicing in the following vowel (the onset of regular pitch pulses in the waveform).

### **Q.Lack of invariance**

'lack of phonetic invariance' provides us with many reasonable justifications as it has posed an important problem for phonetic theory as we try to reconcile the fact that shared phonetic knowledge can be described using the IPA symbols and phonological features with the fact that the individual phonetic forms that speakers produce and hear on a daily basis span a very great range (of varieties). This lack of invariance as a problem also has great practical significance for language engineers who try to get computers to produce and recognize speech.

### **Q.What do the first two formant frequencies tell about vowel?**

The first two frequencies are important here. The first formant (F1) is inversely related to the height of a vowel whereas the second formant (F2) is related to the frontness of a vowel sound. When the first two formants are taken,

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the vowels of a language can be plotted on a chart and the structure is very much related to the traditional description of vowel sounds.

## Q. Duration of vowel preceding a consonant.

There are three important acoustic correlates of voicing in stops: the voice bar, VOT, and the duration of the preceding vowel. Record /apa/, /aba/, /ata/, /ada/, /apha/ and /atha/ and for each of the stops in the file, take the three measurements according to the following instructions: See the voicing or the voice bar by exploring features of stop. We can also explore the features related to the place of articulation (any bilabial feature for /p/ or /b/ in comparison with non-bilabial). Also check the duration of the preceding vowels. Note down the presence of voicing.

## Q. How to create a text-grid? 5

Create a textgrid:

- In the Praat Objects window, highlight the subject (required) file.
- Annotate > To TextGrid.
- Create two tiers (this will be enough for our purposes). Write 'word segment' (these are two tiers) on the cell named 'All tier names' on the small window

## Q. dental stop.....5

Dental Sounds are present both in British and American English, e.g. dental fricatives [θ, ð] but there are no dental stops, nasals, or laterals except allophonically realized (before [θ, ð] as in eighth, tenth, wealth). Many speakers of French, Italian, and other languages (such as Urdu, Pashto and Sindhi) typically have dental stops such as [t̪, d̪]. However, there is a great deal of individual variation in the pronunciation of these consonants in all these languages.

## Q. feature hierarchy any three.....3

Stop, Fricative, Approximant, and Vowel

## Q. Japanese as syllable timed language .....3

A classic example is Japanese in which all morae have approximately the same duration. This tendency is contrasted with stress-timing where the time between stressed syllables is said to tend to be equal irrespective of the number of unstressed syllables in between.

## Q. Name following

1. [f,v] Labiodental fricatives
2. [P,b,m] Stops

## Q. two things taken by spectral slice .....2

Spectral slices (or cross-sections) show the amplitude/frequency spectrum at a selected moment in the signal. They are useful as aids to comparing local spectral events or measuring spectral properties such as formant frequencies, levels or bandwidths.

## Q. One phoneme pattern

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One phoneme pattern V I, oh, owe

## Q. Two phoneme pattern

Two phoneme pattern CV to, be, see

## Q. Three phoneme pattern ,example from English?

Three phoneme pattern CVC cat, dog, run

## Q. phonetics OF THE COMMUNITY?

Linguistic phonetic descriptions (of speech sounds) are, by and large, descriptions of the phonetics of the community (excluding the individual properties and considering the shared properties of the sound system of a language by its native speakers). The representations that experts write and use in the IPA, and analyze in a formal phonological theory, are intended to show the community's shared knowledge of how to say the words of a language. It is important to note that this shared phonetic knowledge is perceptible to other speakers (and thus to the phonetician as well) and experts are mainly related to the aggregate behavior of linguistic group in the sense that it captures what community members accept as correct pronunciation system.

## Q. Phonology fields?

Phonological studies can be divided into two fields: segmental phonology and suprasegmental phonology.

## Q. keeping the view of tongue and lips describe ( i,e,ɛ,a)?

part of the tongue - the front, the middle or the back of the tongue may be raised, giving different vowel qualities: compare /æ/ vowel (as in word 'cat') as a front vowel, with the /ɑ:/ vowel (as in 'cart') which is a back vowel. Thirdly, the tongue (and the lower jaw) may be raised 'close' to the roof of the mouth (for close vowels. e.g. /i:/ or /u:/), or the tongue may be left 'low' in the mouth with the jaw comparatively 'open' (as for open vowels e.g., /a:/ and /æ/).

## Q. coronal three types

Coronal' articulations can be split into three mutually exclusive possibilities: Laminal (i.e., blade of the tongue), Apical (i.e., tip of the tongue), and Sub-apical (i.e., the under part of the blade of the tongue)

## Q. Explain with reference to the division of sound into supra laryngeal n lyrangeal. 5

Sounds are divided in terms of their supra-laryngeal and laryngeal characteristics, and their airstream mechanism. The supra-laryngeal characteristics can be further divided into those for place (of articulation), manner (of articulation), the possibility of nasality, and the possibility of being lateral. Thus, these features are used for classifying speech sounds and describing them formally.

## Q: how many syllables in word minimization, seperate with hyphens. 2

/mɪ. nɪ. maɪ. zeɪ. ʃ(ə)n/

CV.CV.CV.CV.CVC

## Q: which phonetic branch is dependent on the use of instruments? 2

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The latest trends under experimental phonetics include brain functions in speech production and processing (by using the latest equipment – many special instruments such as x-ray techniques)

## **Q: describe the cardinal vowels according to lips, tongue and jaw position. /æ/ and /ə/ (3 marks)**

Firstly, the shape of the lips (lip-rounding), rounded (for sounds like /u:/ vowel), neutral (as for ə - schwa sound) or spread (as in /i:/ sound in word like sea or – when photographers traditionally ask you to say “cheese” /tʃi:z/ in order to make you look smiling. Secondly, part of the tongue - the front, the middle or the back of the tongue may be raised, giving different vowel qualities: compare /æ/ vowel (as in word ‘cat’) as a front vowel, with the /ɑ:/ vowel (as in ‘cart’) which is a back vowel. Thirdly, the tongue (and the lower jaw) may be raised ‘close’ to the roof of the mouth (for close vowels. e.g. /i:/ or /u:/), or the tongue may be left ‘low’ in the mouth with the jaw comparatively ‘open’ (as for open vowels e.g., /a:/ and /æ/).

## **Q. Relationship between Formants and Harmonics:**

In the last two topics, we measured the harmonics as well as the formants of the sonorant sounds (vowels). Having taken the measurements for both formants and harmonics, we need to compare them and explore a possible relationship between the two. Captured in the source-filter model of the speech, it is clear now (from the comparison of the two values – for formants and harmonics) that harmonic numbers are different for one type of sounds but the formants are the same. The relationship between the harmonics and the formants is captured in the source-filter model of speech production. The point is that harmonics are related to the laryngeal activity (source) and formants are the output of the vocal tract (filter).

## **Q. steps to follow harmonics (5)**

- Go to: Spectrum > Spectrogram settings.
- Change the window length to 0.025s – the default window length is 0.005s (wide-band spectrogram) - this changes the spectrogram dramatically!
- Looking at each vowel, notice the grey horizontal bands: these correspond to harmonics. For each vowel, measure the frequencies of the first 3 harmonics (H1-H3) and the 10th harmonic (H10).
- Click on the center (horizontally) of each harmonic in the center of each vowel.
- A red horizontal bar should appear with the frequency value on the left side of the window in red.

## **Q. 3 types of vot**

voiced, voiceless and a voiceless aspirated sound.

## **Q. japanese as syllable**

A classic example is Japanese in which all morae have approximately the same duration. This tendency is contrasted with stress-timing where the time between stressed syllables is said to tend to be equal irrespective of the number of unstressed syllables in between.

## **Q. vot of /p/ and/ph/**

In a voiceless unaspirated plosive (such as /p/ there is a delay (or lag) before voicing starts; and, in a voiceless aspirated plosive (e.g., /p<sup>h</sup>/), the delay is much longer, depending on the amount of aspiration. The amount of this delay is called Voice Onset Time (VOT) which in relation to the types of plosive varies from language to language.

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## Q. What is praat?

Nowadays most of the research works in phonetics and phonology are based on software like Praat and WaveSurfer. So it is appropriate to include some beginners' level introductory sessions to one of the mostly used software Praat.

Praat is a computer program with which you can analyze, synthesize, and manipulate speech, and create high-quality pictures for your articles and thesis.

## Q. Mechanism of source filter ?

In this theory, the tract is represented using a source-filter model and several devices have been devised to synthesize speech in this way. The idea is that the air in the vocal tract acts like the air in an organ pipe, or in a bottle. Sound travels from a noise-making source (i.e., the vocal fold vibration) to the lips. Then, at the lips, most of the sound energy radiates away from the lips for a listener to hear, while some of the sound energy reflects back into the vocal tract. The addition of the reflected sound energy with the source energy tends to amplify energy at some frequencies and damp energy at others, depending on the length and shape of the vocal tract. The vocal folds (at larynx) are then a source of sound energy, and the cavity (vocal tract - due to the interaction of the reflected sound waves in it) is a frequency filter altering the timbre of the vocal fold sound. This idea can make it very easy for us to understand the formants of a vowel sound. Thus this same source-filter mechanism is at work in many musical instruments. In the brass instruments, for example, the noise source is the vibrating lips in the mouthpiece of the instrument, and the filter is provided by the long brass tube.

## Q. Three syllable structure example

Syllable structure could be of three types: 'simple' (CV), 'moderate' (CVC) and 'complex' (with consonant clusters at edges) such as CCVCC and CCCVCC (where V means vowel and C stands for consonant).

## Q. Why air vibrate more than in one way in vocal tract?

Any particle of air, such as that in the vocal tract or that in a bottle, will vibrate in a way that depends on its size and shape. Remember that the air in the vocal tract is set in vibration by the action of the vocal folds (in larynx). Every time the vocal folds open and close, there is a pulse of acoustic energy (activation). Irrespective of the rate of vibration at source (of the vocal folds), the air in the vocal tract will resonate at these frequencies as long as the position of the vocal organs remains the same. Because of the complex shape of the filter (tract), the air will vibrate in more than one way at once.

## Q. Two rhythmic patterns in English Language

1. stress timed language and
2. syllable timed languages.

## Q. Why we have to focus more on volume bar while recording in Praat?

Make sure the volume bar is fluctuating as you record – if it isn't, you're not recording; if you don't see the volume bar at all, you're not speaking loudly enough.

- Watch out for clipping. If your recording level is too high and you go into the red on the volume bar, you'll end up with what is called a "clipped" signal; this is very bad for speech analysis!

## Q. What is rhoticization? Give examples.

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In the description of vowel quality, rhotacization (or rhotacized vowel) is a term which is used in English phonology referring to dialects or accents where /r/ is pronounced following a vowel, as in words 'car' and 'cart'. Thus varieties of English are divided on the basis of this feature - varieties having this feature are rhotic (in which /r/ is found in all phonological contexts) while others (not having this feature) are non-rhotic (such as Received Pronunciation where /r/ is only found before vowels as in 'red' and 'around'). Similarly, vowels which occur after retroflex consonants are sometimes called rhotacized vowels (they display rhotacization).

### Q. Semivowels

Most of the world languages contain a class of sounds that functions in a way similar to consonants but is phonetically similar to vowels (e.g., in English, /w/ and /j/ as in 'wet' and 'yet'). When they are used in the first part of syllables (at onset), they function as consonants. But if they are pronounced slowly, they resemble (in quality) with the vowels [u] and [i] respectively. These sounds are called semivowels which are also termed as approximants today. In French there are three semivowels (i.e., in addition to j and w there is another sound symbolized /ɥ/ and is found in initial position in the word like 'huit' /ɥit/ (eight) and in consonant clusters such as /frɥ/ in /frɥi/ ('fruit'). The IPA chart also lists a semivowel corresponding to the back close unrounded vowel /u/. Like the others, this is classed as an approximant.

### Q. Palatalization

'palatalization' (adding a high front tongue gesture as in sound /i/)

Palatalization is the addition of a high front tongue gesture (as in sound like [i]) to another main (primary) gesture. The diacritic used for palatalization is the small [j] superimposed above another symbol (for primary gesture). The terms palatalization (a process whereby the place of an articulation is shifted nearer to the center of the hard palate) and palatalized (when the front of the tongue is raised close to the palate while an articulatory closure is made at another point in the vocal tract) are sometimes used in a slightly different way. A palatalized consonant has a typical /j/-like (similar to /i/ vowel) quality.

### Q. Measuring the harmonics:

Harmonics are the multiple integers of the fundamental frequency which are basically the result of vocal fold vibration (complex wave). We need the 'narrow band spectrogram' for measuring the H (which we can set by fixing the spectrum setting at 0.025). By starting measuring the frequency of the first three harmonics, we will go to the H10 (H1, H2, H3 – H10). Finally, we will compare with pitch measurement already taken. It is important to note down that when our vocal folds vibrate, the result is a complex wave, consisting of the fundamental frequency (which you have measured in Topic 187) plus other higher frequencies, called harmonics. As already mentioned, to see harmonics, we need to look at a narrow-band spectrogram, which is more precise along the frequency domain than the default wide-band spectrogram. Let's now take the harmonics:

#### 1. Display a narrow-band spectrogram:

- Go to: Spectrum > Spectrogram settings.
- Change the window length to 0.025s – the default window length is 0.005s (wide-band spectrogram) - this changes the spectrogram dramatically!
- Looking at each vowel, notice the grey horizontal bands: these correspond to harmonics. For each vowel, measure the frequencies of the first 3 harmonics (H1-H3) and the 10th harmonic (H10).
- Click on the center (horizontally) of each harmonic in the center of each vowel.

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- A red horizontal bar should appear with the frequency value on the left side of the window in red.

## Q. Ease of articulation

In order to explain the sound patterns of a language, the views of both speaker and listener are considered. Both of them like to use the least possible articulatory effort (except when they are trying to produce very clear speech), and there are a large number of assimilations, with some segments left out, and other reduced to minimum. Thus a speaker uses language with an ease of articulation (e.g., coarticulation and secondary articulation). This tendency to use language sounds with maximum possible ease of articulation leads to change in the pronunciation of words.

## Q. features hierarchy

Feature hierarchy is an important concept in phonetics and phonology which is based on the properties and features of sounds. In a very general sense, a feature may be tied to a particular articulatory maneuver or acoustic property. For example, the feature [bilabial] indicates not only that the segment is produced with lips but also that it involves both of them. Such features (in phonetics and phonology) are listed in a hierarchy with nodes in the hierarchy defining ever more specific phonetic properties. For example, sounds are divided in terms of their supra-laryngeal and laryngeal characteristics, and their airstream mechanism. The supra-laryngeal characteristics can be further divided into those for place (of articulation), manner (of articulation), the possibility of nasality, and the possibility of being lateral. Thus, these features are used for classifying speech sounds and describing them formally.

## Q. Pitch as suprasegmental feature

As a suprasegmental feature, pitch is an auditory sensation - when we hear a regularly vibrating sound such as a note played on a musical instrument (or a vowel produced by the human voice), we hear a high pitch (when the rate of vibration is high) and a low pitch (when the rate of vibration is low). There are some speech sounds that are voiceless (e.g. /s/), and cannot give rise to a sensation of pitch in this way but the voiced sounds can. Thus the pitch sensation that we receive from a voiced sound corresponds quite closely to the frequency of vibration of the vocal folds. However, we usually refer to the vibration frequency as fundamental frequency in order to keep the two things distinct. In tonal languages, pitch is used as an essential component of the pronunciation of a word and a change of pitch may cause a change in meaning. In most languages (whether or not they are tone languages) pitch plays a central role in intonation. In very simple words, pitch is the variation in the vibration of vocal folds.

## Q. Blank cells of IPA

One evidence that the IPA chart is based on linguistic phonetics is the description of the blank cells on the chart (those neither shaded nor containing a symbol) that indicate the combinations of categories that are humanly possible but have not been observed so far to be distinctive in any language (e.g., a voiceless retroflex lateral fricative is possible but has not been documented so far, so it is left blank). The shaded cells, on the other hand, exhibit the sounds not possible at these places.

## Q. Syllable

In a simple way of defining the term, syllables are the parts of word (in which a word is further divided into parts), for example, mi-ni-mi-za-tion or sup-re-seg-men-tal. Phonetically, we can observe that the flow of speech typically consists of an alternation between vowel-like states (where the vocal tract is comparatively open and unobstructed) and consonant-like states where some obstruction to the airflow is made (thus altering speech between the two natural kinds of sounds). So, from the speech production point of view, a syllable consists of a movement from a constricted or silent state to a vowel-like state and then back to constricted or silent state. From the acoustic point of view, this means that the speech signal shows a series of peaks of energy corresponding to vowel-like states separated by troughs of lower energy (sonority).

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## **Q. Explain stress timed languages**

Languages of the world are; therefore, divided into two broad categories: stress timed language and syllable timed languages. Stress timed languages have stress as their dominating rhythmic feature meaning that these languages seem to be timed according to the stressed patterns (the division among the syllables is made on the basis of stressed and unstressed patterns e.g., English and German languages). In other words, in stress timed languages, stressed syllables occur with regular intervals and their units of timing are perceived accordingly. Stress-timed rhythm is one of these rhythmical types, and is said to be characterized by a tendency for stressed syllables to occur at equal intervals of time.

## **Q. why are the nasals and glides called Sonorants. (2)**

These sounds are called sonorants because they have formants (remember their acoustic correlates).

Sonorants are vowel-like sounds (nasals and glides). These sounds are called sonorants because they have formants. But they are different from vowels because they generally have lower amplitude; therefore, they behave like consonants

## **Q. what specific terms are used for the consonants cluster Cc and ccc in syllable? (2)**

Consonant sequences are called clusters (e.g., CC – two consonants or CCC – three consonants). Most of the phonotactic analyses are based on the syllable structures and syllabic templates.

## **Q. what are three formants help in distinguishing vowel from each other?**

the lowest three formants distinguish vowels from each other

## **Q. what is acoustic correlate of vowel height? (2)**

Acoustically, vowels are mainly distinguished by the first two formant frequencies F1 and F2; F1 is inversely related to the vowel height (which means that smaller F1 amplitude = higher vowels), and F2 is related to the front or back of the vowels (smaller F2 amplitude = more back vowels).

## **Q. while recording on praat, one has to be careful about clipping, explain? (3)**

Watch out for clipping. If your recording level is too high and you go into the red on the volume bar, you'll end up with what is called a "clipped" signal; this is very bad for speech analysis.

## **Q. write a note on structure of syllables? (5)**

Phonologists are interested in the structure of a syllable. It can be divided into three possible parts as phonemes may occur at the beginning (onset), in the middle (nucleus or peak) and at the end (coda) of syllables - the combination of nucleus (peak) and coda is called the rhyme. The beginning (onset) and ending (coda) are optional while a syllable must have a nucleus (at least one phoneme). Thus, the study of the sequences of phonemes is called phonotactics, and it seems that the phonotactic possibilities of a language are determined by its syllabic structure (sequences of sounds that a native speaker produces can be broken down into syllables).

## **Q. discuss the role of memory as presented by exemplar theory?**

The role of the memory for speech under the exemplar theory suggests that many instances of each word are stored in memory and their phonetic variability is memorized rather than computed. No one style is basic (from which others are derived), because all are stored in memory.

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## Q. In order to Formant automatically, what three steps will you follow, elaborate?(5)

- Display the formant track: Formant > Show formants.
- Place your cursor in the middle, stable portion of the vowel.
- Go to Formant > Formant listing: a box will appear with the time point at which the measurement was taken, and the first four formants.

## Q. How to boundary remove in Praat software.

To remove a boundary that you have made - Highlight the boundary - Go to Boundary > Remove OR click Alt+backspace.

## Q. stop voicing

There are three important acoustic correlates of voicing in stops: the voice bar, VOT, and the duration of the preceding vowel. Record /apa/, /aba/, /ata/, /ada/, /apha/ and /atha/ and for each of the stops in the file, take the three measurements. See the voicing or the voice bar by exploring features of stop. Also check the duration of the preceding vowels. Note down the presence of voicing.

## Q. Explain features of SAGS:

1. Palatalization [j] is the raising of the front of the tongue such as for /i/ vowel).
2. Velarization (written as [v] and [ɹ]) is the raising of the back of the tongue (such as [u]-like sound.
3. Pharyngealization [ʕ] is the retracting of the root of the tongue.
4. Labialization [w] is the rounding of the lips such as Arabic [s<sup>w</sup>] and [t<sup>w</sup>].

## Q. Define Syllable

In a simple way of defining the term, syllables are the parts of word (in which a word is further divided into parts), for example, mi-ni-mi-za-tion or sup-re-seg-men-tal. Phonetically, we can observe that the flow of speech typically consists of an alternation between vowel-like states (where the vocal tract is comparatively open and unobstructed) and consonant-like states where some obstruction to the airflow is made (thus altering speech between the two natural kinds of sounds). So, from the speech production point of view, a syllable consists of a movement from a constricted or silent state to a vowel-like state and then back to constricted or silent state. From the acoustic point of view, this means that the speech signal shows a series of peaks of energy corresponding to vowel-like states separated by troughs of lower energy (sonority).

## Q. Explain IPA charts

One evidence that the IPA chart is based on linguistic phonetics is the description of the blank cells on the chart (those neither shaded nor containing a symbol) that indicate the combinations of categories that are humanly possible but have not been observed so far to be distinctive in any language (e.g., a voiceless retroflex lateral fricative is possible but has not been documented so far, so it is left blank). The shaded cells, on the other hand, exhibit the sounds not possible at these places. Further, below the consonant chart is a set of symbols for consonants made with different airstream mechanisms (clicks, voiced implosives, and ejectives). All these descriptions reflect the potentialities of human speech sounds (as a linguistic community) not only showing the possible segments but also the suprasegmental features and points related to the possible airstream mechanisms and even the diacritics for

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various types of coarticulations and secondary articulatory gestures. The IPA chart is carefully documented (by experts) and is continuously revised and updated.

## **Q. Experimental Techniques about phonetics**

In experimental phonetics and phonology, the studies of sounds include various latest experimental techniques and computer software that are used under carefully designed lab experimentation. It is an important aspect of the application of the latest technology by going beyond the simple acoustics and by working in sophisticated phonetic labs in order to discover the hidden aspects of human speech. For example, questions such as 'How speech is produced and processed?' are the focus of experimental phonetics (explore the speech chain as the beginning of experimental phonetics as mentioned in Chapter-20 by Peter Roach). The latest trends under experimental phonetics include brain functions in speech production and processing (by using the latest equipment – many special instruments such as x-ray techniques), speech errors, neurolinguistics and the topics related to the developments through computers – for speech analysis and synthesis.

## **Q. Importance of Spectrograms**

1. Using Praat (or any other software) and spectrogram is particularly useful when a researcher is working on a problem related to the nature (physical properties) of a sound (e.g., is it a phoneme or allophone?).
2. It increases our understanding of the speech sounds and their behavior in different forms (in isolation or as the part of connected speech).
3. Practice on spectrogram gives us the opportunity to learn about the characteristics of speech sounds.
4. It is also important for experts who are working on phonetic aspects of speech as signal processing.
5. These are also used as the part of techniques in speech recognition.

## **Q. 2 formants f1 and f2:**

Formants are the overtone resonances. Acoustically, in order to plot vowels on chart, F1-F2 are very important. We need the wide bands for measuring the formants (which are the important characteristics of sonorant speech sounds – vowels). On spectrogram, formants are thick bands (darkness corresponds to loudness; i.e. the darkest harmonics are the ones that are the most amplified). These amplified harmonics form the formants that are characteristic of sonorant speech sounds.

## **Q. define rhotacized how it is produced a sound.**

In the description of vowel quality, rhotacization (or rhotacized vowel) is a term which is used in English phonology referring to dialects or accents where /r/ is pronounced following a vowel, as in words 'car' and 'cart'. Thus varieties of English are divided on the basis of this feature - varieties having this feature are rhotic (in which /r/ is found in all phonological contexts) while others (not having this feature) are non-rhotic (such as Received Pronunciation where /r/ is only found before vowels as in 'red' and 'around'). Similarly, vowels which occur after retroflex consonants are sometimes called rhotacized vowels (they display rhotacization). It is important to mention that while BBC pronunciation is nonrhotic, many accents of the British Isles are rhotic, including most of the south and west of England, much of Wales, and all of Scotland and Ireland. Most American English speakers speak with a rhotic accent, but there are non-rhotic areas (e.g., the Boston area, lower-class of New York and the Deep South).

## **Q. provide word which change to bilabial to labiodental. (RECHECK THIS ON YOUR OWN)**

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very common in English (e.g., stops and nasal: p, b, m), In some languages (such as Ewe of West Africa), bilabial fricatives contrast with labiodental fricatives. The symbols for the voiceless and voiced bilabial fricatives are [ɸβ]. These sounds are pronounced by bringing the two lips nearly together, so that there is only a slit between them. Ewe also contrasts voiceless bilabial and labiodental fricatives.

## Q. two main categories of language

Languages of the world are; divided into two broad categories: stress timed language and syllable timed languages. Stress timed languages have stress as their dominating rhythmic feature meaning that these languages seem to be timed according to the stressed patterns(the division among the syllables is made on the basis of stressed and unstressed patternse.g., English and German languages). In other words, in stress timed languages, stressed syllables occur with regular intervals and their units of timing are perceived accordingly. Stress-timed rhythm is one of these rhythmical types, and is said to be characterized by a tendency for stressed syllables to occur at equal intervals of time.

## Q. how stress time language change to syllabic time language.

‘Stress timed languages’ is a very general phrase used in phonetics to characterize the pronunciation of languages displaying a particular type of rhythmic pattern that is opposed to that of syllable-timed languages. In stress-timed languages, it is claimed that the stressed syllables recur at regular intervals of time (stress-timing) regardless of the number of intervening unstressed syllables as in English. This characteristic is sometimes also referred to as ‘isochronism’, or isochrony. However, it is clear that this regularity is the case only under certain conditions, and the extent to which the tendency towards regularity in English is similar to that in, say, other Germanic languages remains unclear. In short, the division among the syllables is made on the basis of stress and unstressed patterns. In such languages, stress is realized both at word and sentence levels approximately changing the rhythmic patterns (particularly at sentence level).

## Q. tube model. (5)

the formants that characterize different vowels are the result of the different shapes of the vocal tract. Any particle of air, such as that in the vocal tract or that in a bottle, will vibrate in a way that depends on its size and shape. Remember that the air in the vocal tract is set in vibration by the action of the vocal folds (in larynx). Every time the vocal folds open and close, there is a pulse of acoustic energy (activation). Irrespective of the rate of vibration at source (of the vocal folds), the air in the vocal tract will resonate at these frequencies as long as the position of the vocal organs remains the same. Because of the complex shape of the filter (tract), the air will vibrate in more than one way at once. So, the relationship between resonant frequencies and vocal tract shape is actually much more complicated than the air in the back part of the vocal tract vibrating in one way and the air in other parts vibrating in another. Here we will just remember the fact that in most voiced sounds, three formants are produced every time the vocal folds (source) vibrate. Note an interesting point here that the resonance in the vocal tract (filter) is independent of the rate of vibration of the vocal folds (source). In other words, the vocal folds may vibrate faster or slower, giving the sound a higher or lower pitch, but the formants will be the same as long as the position of the tube (vocal tract) is the same.

## Q. three steps labeling on praat (5)

### 1. Create a textgrid:

- In the Praat Objects window, highlight the subject (required) file.
- Annotate > To TextGrid.

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- Create two tiers (this will be enough for our purposes). Write ‘word segment’ (these are two tiers) on the cell named ‘All tier names’ on the small window.

## 2. Open the sound file and textgrid together:

- Hold down Ctrl and click on each file to highlight them both.
- Edit (in your display you should now see the waveform (top), the spectrogram (middle) and the textgrid (bottom) corresponding to your sound file).

## 3. Segment the file:

- Place the cursor at the beginning of the name on the spectrogram/waveform; a boundary line will show up.
- Click in the little circle at the top of the word tier in the Textgrid to create a boundary.
- To remove a boundary that you have made - Highlight the boundary - Go to Boundary > Remove OR click Alt+backspace.

## Q. Glides

Glides are also the sonorants (vowel-like) sounds as they have similar patterns (have formants). Take the first three formants (F1, F2 and F3) from the middle of the sounds for glides (both for /w/ and /j/) and explore their acoustic correlates. Carefully judge the center of these sounds (the midpoint of [w] and [j]). Analyze that how similar is the formant structure of glides with vowels and nasals. Draw lines to indicate F1, F2, F3 and compare with vowels.

## Q. cardinal vowels:

In order to classify vowels (independent of the vowel system of a particular language), the English phonetician Daniel Jones introduced a system in early 20th century and worked out on a set of vowels called the —cardinal vowels comprising of eight vowels to be used as reference points (so that other vowels could be related to them like the corners and sides of a map). Cardinal vowel system is a chart or four-sided figure (the exact shape of which has been changed from time to time), with eight corners as can be seen on the IPA chart from IPA website. It is a diagram to be used both for rounded and unrounded vowels, and Jones proposed that there should be a primary and a secondary set of cardinal vowels. The primary set includes eight vowels in total (from 1 to 8 ; the front unrounded vowels [i, e, ε, a], the back unrounded vowel [ɑ] and the rounded back vowels [ɔ, o, u].

## Q. In spectrograms, what area indicates the intensity of components? 2

In spectrograms, time runs from left to right, the frequency of the components is shown on the vertical scale, and the intensity of each component is shown by the degree of darkness. It is thus a display that shows, roughly speaking, dark bands for concentrations of energy at particular frequencies—showing the source and filter characteristics of speech.

## Q. Which varieties of English are called rhotic? 2

Varieties having this feature are rhotic (in which /r/ is found in all phonological contexts)

## Q. How tone (high vs low) change the meaning of the word [ma] in Mandarin? 2

For example, in Mandarin Chinese, [ma] said with a high pitch means ‘mother’ while [ma] said on a low rising tone means ‘hemp’. In other (non-tonal) languages, tone forms the central part of intonation, and the difference between, for example, a rising and a falling tone on a particular word may cause a different interpretation of the sentence in

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which it occurs. In the case of tone languages, it is usual to identify tones as being a property of individual syllables, whereas an intonational tone may be spread over many syllables.

### **Q. In what way Voice Onset time is related to the aspirated sounds? 2**

In order to understand VOT, the three types of plosive sounds are to be explained – voiced, voiceless and a voiceless aspirated sound. Most aspirated (largest positive VOT) at the top to most voiced (largest negative VOT) at the bottom. The Navajo aspirated stops have a very large VOT value that is quite exceptional (150 MS).

### **Q. Why the English lateral /l/ is called an approximant? 3**

#### **Or Difference of lateral sounds of Americans and British English?**

The only English lateral phoneme, at least in British English, is /l/ with allophones [l] as in led [lɛd] and [ɫ] as in bell [bɛɫ]. In most forms of American English, initial [l] has more velarization than is typically heard in British English initial [l]. In all forms of English, the air flows freely without audible friction, making this sound a voiced alveolar lateral approximant. It may be compared with the sound [ɭ] in red [ɹɛd], which is for many people a voiced alveolar central approximant. Laterals are usually presumed to be voiced approximants unless a specific statement to the contrary is made.

### **Q. name some materials that can assist in the teaching of the language skills. 5**

1. Explore already developed material available online from various sources (such as British Council and other teacher resource centers); however, you must also be able to develop your own material (as specifically required by your students).
2. You can develop your material related to the pronunciation teaching to the learners of English.
3. You can incorporate material related to the IPA text – transcription of the audio (listening) based activities – by involving students on using dictionaries (ideally the phonetic dictionaries) in the classroom.
4. Movies and documentaries (such as from BBC - CNN - National Geographic channels) may also serve as very effective resources for the teaching of pronunciation.
5. Finally, the real life material (for listening) and writing interaction from everyday language may also yield tremendous results. The focus of material development should always be the enhancement of the proficiency level of students.

### **Q. describe two reasons why phonetics of the community is considered for phonetic description.5**

Firstly, individual speakers differ in interesting ways (two native speakers of a language will always speak with some variations). The description of the phonetics of the individual involves describing the phonetic knowledge and skills related to the performance of language. It is possible that certain aspects of the phonetics of the individual can be captured using IPA transcription but others are not compatible with it (such as his private knowledge and its performance and the role of memory and experience).

Secondly, the phonetics of the individual is usually not the focus of the linguist in speech elicitation, and it is difficult to describe even with spectrograms of the person's speech. Although, the phonetics of the individual is the focus of much of the explanatory power of phonetic theory but for general phonetic description we need to focus on the phonetics of the community.

### **Q. briefly discuss lexical stress.**

In terms of its linguistic function, stress is often treated under two different headings: word (lexical) stress and sentence (emphatic) stress. Lexical stress is basically related to the primary stress applied at syllable level (when

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only one syllable is stressed) that has the ability to change the meaning and the grammatical category of a word as in the case of im port' (noun and im ORT' (verb).

## Q. how are the oral stops produced. Provide IPA symbols for eng oral stops any three. 5

In phonetics, a **stop**, also known as a **plosive** or **oral occlusive**, is a consonant in which the vocal tract is blocked so that all airflow ceases.

The occlusion may be made with the tongue blade ([t], [d]) tongue body ([k], [g]), lips ([p], [b]), or glottis ([ʔ]). Stops contrast with nasals, where the vocal tract is blocked but airflow continues through the nose, as in /m/ and /n/, and with fricatives, where partial occlusion impedes but does not block airflow in the vocal tract.

## Q. Why do you think teacher are mostly expected to perform action research in ELT. 3

Teachers are expected to facilitate action research which is the most rewarding and productive for their own profession. For example, the phonetics of phonological speech errors if explored and shared by teachers (by investigating their own practices) may lead to a very positive discussion in the academic circles (of research into ELT – SLA). Similarly, topics such as learners' performance and development (e.g., what do good speakers do?) may yield useful results for teachers' fraternity. Having said this, it is required from teachers (and student teachers) to facilitate action research related to reading/listening issues, English reading strategies (e.g., in primary schools) – (and their effectiveness), impact on pronunciation and many more. Research in the fields of phonetic theory and the description with phonological, typological and broader implication may also be included in phonetics and phonology specific action research.

## Q. three dimensions of stress. 3

there are three dimensions:

1. fixed word stress (mainly found in Romance languages),
2. variable word stress (mainly found in languages such as English and German) and
3. fixed phrase stress (phrase as a third possibility as exhibited by Japanese) and they want to categorize languages on the basis of these three patterns

## Q. Phonotactics

In phonology, phonotactics is the study of the ways in which phonemes are allowed to combine in a particular language. (A phoneme is the smallest unit of sound capable of conveying a distinct meaning.) Over time, a language may undergo phonotactic variation and change. For example, as Daniel Schreier points out, "Old English phonotactics admitted a variety of consonantal sequences that are no longer found in contemporary varieties"

## Q. in what way sonorants are different from vowels. 2

They are different from vowels because they generally have lower amplitude; therefore, they behave like consonants. Record the following sequences for our experimentation on sonorant sounds /ama/ - /ana/ - /aŋa/ - /wi/ - /ju/. Having recorded these sequences, now start exploring the features of these sounds like the measurement of F1, F2, F3 and also try to compare them with vowels.

## Q. define sound change as one of the postulate of yhe exemplar theory. 2

Sound change is phonetically gradual and operates across the whole lexicon. It is a gradual shift as new instances keep on adding.

## Q. on what basis Akan (language in Ghana) constraintsits vowels?2

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One out of such variations is advanced tongue root (ATR) which is found in Akan language spoken in Ghana. Actually, vowels produced with ATR involve the furthest-back part of the tongue, opposite to the pharyngeal wall, which is not normally involved in the production of speech sounds - also called the radix (articulations of this type may, therefore, be described as radical). ATR (a kind of articulation in which the movement of the root of tongue expands the front-back diameter of the pharynx) is used phonologically in Akan (and some other African languages) as a factor in contrast of vowel harmony. The opposite direction of movement is called retracted tongue root (RTR). ATR is thus related to the size of pharynx – making the pharyngeal cavity different: creating comparatively large (+ATR: root forward and larynx lowered) and small pharyngeal cavity (-ATR: no advanced tongue root). Akan contrasts between two sets of vowels +ATR and –ATR.

### **Q. Which varieties of English are called non-rhotic? 3**

Non-rhotic (such as Received Pronunciation where /r/ is only found before vowels as in ‘red’ and ‘around’).

Most American English speakers speak with a rhotic accent, but there are non-rhotic areas (e.g., the Boston area, lower-class of New York and the Deep South).

### **Q. Write three variables of pitch. 3**

Three variables of pitch

1. Pitch range,
2. height and
3. direction

### **Q. Write five words where consonants are clustered. 5**

cry  
create  
crazy

price  
practice  
private

### **Q. Name the five major features based on the major regions of vocal tract. 5**

The five features in total (i.e., Labial, Coronal, Dorsal, Radical, and Glottal).

The first three of these features are related to tongue position whereas ‘Radical’ is a cover term for [pharyngeal] and [epiglottal] articulations made with the root of the tongue.

The feature of ‘Glottal’, on the other hand, is based on being [glottal], to cover various articulations such as [h]. If we are to have a convenient grouping of the features for consonants, we have to recognize that SupraLaryngeal features must allow for the dual nature of the actions of the larynx and include Glottal as a place of articulation. Remember that a sound may be articulated at more than one of the regions Labial, Coronal, Dorsal, Radical, and Glottal.

Within the five general regions, ‘Coronal’ articulations can be split into three mutually exclusive possibilities: Laminal (i.e., blade of the tongue), Apical (i.e., tip of the tongue), and Sub-apical (i.e., the under part of the blade of the tongue). Thus the major regions may be subdivided into sub regions on the basis of their features

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## Q. Define the term "supra-segmental. 5

Supra means 'above' or 'beyond' and segments are sounds (phonemes). Suprasegmental is a term used in phonetics and phonology to refer to a vocal effect (such as tone, intonation, stress, etc.) which extends over more than one sound (segment) in an utterance.

Major suprasegmental features include pitch, stress, tone, intonation or juncture. Phonological studies can be divided into two fields: segmental phonology and suprasegmental phonology.

## Q. VOT (5)

It is the characteristic of voiced + voiceless + aspirates stop sounds and there are very easy steps to calculate the VOT. Record /apa/, /aba/, /ata/, /ada/, /apha/ and /atha/. Zoom in through your stop sounds so that you can analyze the patterns of the stop sounds and find the difference among the three types of VOT (negative, zero and positive). Measure the VOT of each stop and compare voiced/voiceless counterparts (p/b, t/d, k/g). Similarly, zoom in so that you can clearly see the stop closure followed by the beginning of the vowel. You can measure the time between the end of the stop closure (the beginning of the release burst) and the onset of voicing in the following vowel (the onset of regular pitch pulses in the waveform). This is voice onset time or VOT.

## Q. Grammatical Function of intonation. (5)

Its most important function is to act as a signal of grammatical structure (e.g., creating patterns to distinguish among grammatical categories), where it performs a role similar to punctuation (in written language). It may furnish far more contrasts (for conveying meaning). Intonation also gives an idea about the syntactic boundaries (sentence, clause and phrase level boundaries). It also provides the contrast between some grammatical structures (such as questions and statements). For example, the change in meaning illustrated by 'Are you asking me or telling me?' is regularly signaled by a contrast between rising and falling pitch. Note the role of intonation in sentences like 'He's going, isn't he?' (= I'm asking you) opposed to 'He's going, isn't he!' (= I'm telling you) (These examples are given by Peter Roach).

## Q. Nasal Stop (5)

Like stops, nasal can also occur voiced or voiceless (for example, in Burmese, Ukrainian and French) though in English and other most languages nasals are voiced. As voiceless nasals are comparatively rare, they are symbolized simply by adding the voiceless diacritic [ <sup>h</sup> ] under the symbol for the voiced sound. There are no special symbols for voiceless nasals and it is written as /m<sup>h</sup>/ - a combination of the letter for the voiced bilabial nasal and a diacritic indicating voicelessness.

## Q. IPA Text in ELT class. (3)

Developing relevant material for the teaching of phonetics and phonology is an important task for aspiring teachers of English language. For example, you can develop your material related to the pronunciation teaching to the learners of English. You can incorporate material related to the IPA text – transcription of the audio (listening) based activities – by involving students on using dictionaries (ideally the phonetic dictionaries) in the classroom.

## Q. Exemplary Theory. (3)

The role of the memory for speech under the exemplar theory suggests that many instances of each word are stored in memory and their phonetic variability is memorized rather than computed. Exemplar theory says that generalization is also possible within productivity. Interestingly, productivity—the hallmark of linguistic knowledge in the phonetic implementation approach—is the least developed aspect of the exemplar theory.

## **Q. How could I get findings while doing research on Pakistan regional languages? (2)**

Pakistani regional languages are the part of the rich linguistic regions. (Himalaya Hindu Kush (HKH) region, one of the richest regions in the world linguistically and culturally) may be very potential area for research in the fields of areal and typological linguistics (description of linguistic features crosslinguistically). While working on Pakistani regional languages, one may apply for funding from international organizations (e.g., organization for endangered languages and UNISCO).

## **Q. Types of tone. (2)**

Four types of tone is usually used

1. fall,
2. rise,
3. fall–rise and
4. rise–fall

## **Q. ATR stands for / define it.**

Advanced Tongue Root.

ATR : a kind of articulation in which the movement of the root of tongue expands the front–back diameter of the pharynx

## **Q. define sound change (2)**

Sound change is phonetically gradual and operates across the whole lexicon. It is a gradual shift as new instances keep on adding.

## **Q. what does F1 and F2 seems to correspond in term of articulation.**

F1 is inversely related to the vowel height (which means that smaller F1 amplitude = higher vowels)

AND F2 is related to the front or back of the vowels (smaller F2 amplitude = more back vowels.)

## **Q. How can you measure the pitch of vowel. (3)**

There is one more way to confirm your pitch measurement by looking at the spectral slice (which gives the component frequencies and their amplitudes).

- Select a portion of the vowel (70-80ms).
- Spectrum > View spectral slice.
- Click on the first (big) peak = H1 = F0 (Ignore any small spikes at the beginning; this might be noise).

Now note down the frequency of this peak at the top of the vertical bar).

Use the confirmed pitch values and plot the pitch of each vowel on your excel sheet. Make sure you label your y-axis using a scale that allows you to spread out your measurements as much as you can. Now draw the cluster chart from the excel sheet and export to Word document and give the figure number and title.

## **Q. Describe Bilabial gestures. (5)**

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very common in English (e.g., stops and nasal: p, b, m), In some languages (such as Ewe of West Africa), bilabial fricatives contrast with labiodental fricatives. The symbols for the voiceless and voiced bilabial fricatives are [ɸβ]. These sounds are pronounced by bringing the two lips nearly together, so that there is only a slit between them. Ewe also contrasts voiceless bilabial and labiodental fricatives.

### Q. Acceptable and unacceptable sequences

It has often been found that languages do not allow all phonemes to appear in any order (e.g., a native speaker of English can figure out fairly easily that the sequence of phonemes /streŋθs/ makes an English word ('strengths') and that the sequence /bleidg/ would be acceptable as an English word 'blage', although that word does not happen to exist, but the sequence /lvm/ could not possibly be the part of an English word).

### Q. Syllable Timed Languages

In syllable timed languages, all syllables tend to have an equal time value (for example, their length or duration) and the rhythm of the language is said to be syllable-timed. In these languages, syllables tend to occur at regular intervals of time with fixed word stress. A classic example is Japanese in which all morae have approximately the same duration. This tendency is contrasted with stress-timing where the time between stressed syllables is said to tend to be equal irrespective of the number of unstressed syllables in between. Czech, Polish, Swahili and Romance languages (e.g., Spanish and French)

### Q. Explain Velarization. (5)

Velarization involves raising the back of the tongue (adding the /u/ vowel like quality). It can be considered as the addition of an [u]-like tongue position (but remember that it is without the addition of the lip rounding). A typical English example of velarization is the /l/ sound at the end of a syllable (as in words like kill, pill, sell and will) called velarized or dark /l/ and may be written as [ɫ]. The diacritics for velarization are both [ɰ] and [ˑ].

### Q. Briefly explain "ease of articulation". (5)

Ease of articulation (e.g., coarticulation and secondary articulation). This tendency to use language sounds with maximum possible ease of articulation leads to change in the pronunciation of words. In co-articulations, for example, a change in the place of the nasal and the following stop occurred in words such as improper and impossible before these words came into English through Norman French. In words such as these, the [n] that occurs in the prefix in- (as in intolerable and indecent) has changed to [m]. These changes are even reflected in the spelling. In all this and in many similar historical changes, one or more segments are affected by adjacent segments so that there is an economy of articulation.

### Q. Explain theory of Perturbations.

We have discussed that the vocal tract as a tube with a uniform diameter has simultaneous resonance frequencies—several different pitches at the same time. We have also discussed that these resonance frequencies change in a predictable way when the tube is squeezed at various locations. This means that we can model the acoustics of vowels in terms of perturbations of the uniform tube. For example, when the lips are rounded, the diameter of the vocal tract is smaller at the lips than at other locations in the vocal tract. The theory of perturbation says that with the acoustic effect of constriction at the lips, we can predict the formant frequency differences between rounded and unrounded vowels. Keeping in mind this modification in the size and nature of vocal tract (for specific vowel sounds), we can estimate how this perturbation theory works. So for each formant, there are locations in the vocal tract where constriction will cause the formant frequency to rise, and locations where constriction will cause the frequency to fall.

## Q. How ELT teacher develop the material for phonetics and phonology

Developing relevant material for the teaching of phonetics and phonology is an important task for aspiring teachers of English language. The specific needs of ELT activities in one's own context and explore already developed material available online from various sources (such as British Council and other teacher resource centers); however, one must also be able to develop own material (as specifically required by students). For example, the teacher can develop material related to the pronunciation teaching to the learners of English. She/ he can incorporate material related to the IPA text – transcription of the audio (listening) based activities – by involving students on using dictionaries (ideally the phonetic dictionaries) in the classroom. Movies and documentaries (such as from BBC - CNN - National Geographic channels) may also serve as very effective resources for the teaching of pronunciation. Finally, the real life material (for listening) and writing interaction from everyday language may also yield tremendous results. The focus of material development should always be the enhancement of the proficiency level of students.

## Q. Trills and Flap. (2)

In the production of trill the articulator is set in motion by the current of air [r]. It is a typical sound of Scottish English as in words like 'rye' and 'row'.

Flap is front and back movement of tongue tip at the underside of tongue with curling behind. It is found in abundance in Indo-Aryan (IA) languages [ɾ]. Typical flap sounds found in IA languages is a retroflex sound and the examples are [ɽ], [ɖ] and [ɳ].

## Q. Define acoustic phonetics. (2)

Acoustic Phonetics is the study of detailed physical properties of sound we produce. It generally uses tools which read the changes in air pressure that the sound creates. Each sound has its different sound quality, which depends on the source filter, that is our speech organs. Each sound has its own f<sub>0</sub>, f<sub>1</sub>, f<sub>2</sub> and f<sub>3</sub> (formants), depending on the source modifier. f<sub>0</sub> deals with the fundamental frequency, f<sub>1</sub> gives the information about the pharyngeal cavity, f<sub>2</sub> about the oral cavity and f<sub>3</sub> about the position of the lips while the sound was produced.

## Q. What is Neurolinguistics. (GOOGLED THE ANSWER)

**Neurolinguistics** is the study of the neural mechanisms in the human brain that control the comprehension, production, and acquisition of language. As an interdisciplinary field, neurolinguistics draws methods and theories from fields such as neuroscience, linguistics, cognitive science, communication disorders and neuropsychology. Researchers are drawn to the field from a variety of backgrounds, bringing along a variety of experimental techniques as well as widely varying theoretical perspectives. Much work in neurolinguistics is informed by models in psycholinguistics and theoretical linguistics, and is focused on investigating how the brain can implement the processes that theoretical and psycholinguistics propose are necessary in producing and comprehending language. Neurolinguists study the physiological mechanisms by which the brain processes information related to language, and evaluate linguistic and psycholinguistic theories, using aphasiology, brain imaging, electrophysiology, and computer modeling.

## Q.Role of vocal folds and vocal tract in Source filter theory.

In this theory, the tract is represented using a source-filter model and several devices have been devised to synthesize speech in this way. The idea is that the air in the vocal tract acts like the air in an organ pipe, or in a bottle. Sound travels from a noise-making source (i.e., the vocal fold vibration) to the lips. Then, at the lips, most of

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the sound energy radiates away from the lips for a listener to hear, while some of the sound energy reflects back into the vocal tract. The addition of the reflected sound energy with the source energy tends to amplify energy at some frequencies and damp energy at others, depending on the length and shape of the vocal tract. The vocal folds (at larynx) are then a source of sound energy, and the cavity (vocal tract - due to the interaction of the reflected sound waves in it) is a frequency filter altering the timbre of the vocal fold sound. Thus this same source-filter mechanism is at work in many musical instruments. In the brass instruments, for example, the noise source is the vibrating lips in the mouthpiece of the instrument, and the filter is provided by the long brass tube.

### **Q. Explain Intonation in simple words.**

Intonation refers (very) simply to the variations in the pitch of a speaker's voice (f0) used to convey or alter meaning but in its broader and more popular sense intonation covers much of the same field as 'prosody' where variations in such things as voice quality, tempo and loudness are included. Intonation as a suprasegmental feature performs several functions in a language. Its most important function is to act as a signal of grammatical structure (e.g., creating patterns to distinguish among grammatical categories), where it performs a role similar to punctuation (in written language). It may furnish far more contrasts (for conveying meaning). Intonation also gives an idea about the syntactic boundaries (sentence, clause and phrase level boundaries).

### **Q. Explain intonation with reference to syntactic and grammar.**

Intonation also gives an idea about the syntactic boundaries (sentence, clause and phrase level boundaries). It also provides the contrast between some grammatical structures (such as questions and statements). For example, the change in meaning illustrated by 'Are you asking me or telling me?' is regularly signaled by a contrast between rising and falling pitch. Note the role of intonation in sentences like 'He's going, isn't he?' (= I'm asking you) opposed to 'He's going, isn't he!' (= I'm telling you) (These examples are given by Peter Roach).

### **Q. When formants are not steady in PRAAT what will you do.**

- Get rid of Praat's formant tracking: Formant > Show formants (unclick).
- Place your cursor in the center of each formant, in the middle of the vowel.
- A red horizontal bar should appear with the frequency value on the left (in red).

### **Q. Write principles of Features analysis.**

Three principles for feature analysis:

1. contrastive function (how it is different),
2. descriptive function (what it is) and
3. classificatory function (based on broader classes of sounds).

Features may also be studied further as a part of language universals and then their role as language specific sub sets.

### **Q. Explain Secondary articulatory gestures.**

'Secondary' articulation is an articulatory gesture with a lesser degree of closure occurring at approximately the same time as another (primary) gesture. It is different than co-articulation which is at the same time and of the same

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value (taking place as an equal level gesture). Thus it is appropriate to consider four types of secondary articulations in conjunction with vowels because they can usually be described as added vowel-like articulations including; 'palatalization' (adding a high front tongue gesture as in sound /i/), velarization (raising of the back of the tongue), pharyngealization (it is the superimposition of the narrowing of the larynx) and labialization (the addition of lip-rounding).

### Q. What is more authentic in auditory and acoustic analysis.

Acoustic Phonetics is the study of detailed physical properties of sound we produce. It generally uses tools which read the changes in air pressure that the sound creates. Each sound has its different sound quality, which depends on the source filter, that is our speech organs. Each sound has its own f0, f1, f2 and f3 (formants), depending on the source modifier. f0 deals with the fundamental frequency, f1 gives the information about the pharyngeal cavity, f2 about the oral cavity and f3 about the position of the lips while the sound was produced. Auditory phonetics is just the other side of the coin for this study. It deals with the study of these articulated sound characteristics from the perception perspective. Auditory phonetics deals with the listener at a broader aspect. The perceived f0 is measured in terms of pitch and calculated in Mel or bark scales.

### Q. How to become a good teacher and a researchers.

Good teachers are expected to be active researchers and therefore busy in updating themselves about the latest researches and teaching methodologies around the world. It is also a pedagogical challenge for teachers to keep themselves updated by exploring pedagogical and technological challenges for ELT experts (in their own contexts and internationally). For example, the aspects of Task Based Learning and Teaching (TBLT) as a golden method for second language acquisition (SLA) may be effective in Pakistani context if explored by ELT practitioners. Teachers as the agents of change and they must be reading research studies and carry out research by and explore their issues and solutions. A good way is to keep reading teachers' digests and journals and participate in the online discussions by teaching associations.

### Q. filter theory

It is a model of speech (e.g., vowel) production. According to this theory, source refers to the waveform of the vibrating larynx. Its spectrum is rich in harmonics, which gradually decrease in amplitude as their frequency increases. The various resonance chambers of the vocal tract, especially the movements of the tongue and lips, act on the laryngeal source in the manner of a filter, reinforcing certain harmonics relative to others. Thus the combination of these two elements (larynx as source and cavity as filter) is known as the source-filter model of speech (e.g., vowel) production.

### Q. articulatory gestures:

1. **Bilabial gesture** – (e.g., stops and nasal: p, b, m). The symbols for the voiceless and voiced bilabial fricatives are [ɸβ]. These sounds are pronounced by bringing the two lips nearly together, so that there is only a slit between them.
2. **Labiodental fricatives** – [f, v]. In English, a labiodental nasal, [ɱ], may occur when /m/ occurs before /f/, as in emphasis or symphony.
3. **Dental** - e.g. dental fricatives [θ, ð] but there are no dental stops, nasals, or laterals except allophonically realized (before [θ, ð] as in eighth, tenth, wealth). any speakers of French, Italian, and other languages (such as Urdu, Pashto and Hindi) typically have dental stops such as [t̪ d̪].

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4. **Alveolar** are very common targets and stops, nasals, and fricatives all occur in English and in many other languages at alveolar as a target of articulatory gestures (e.g., t, d, n, l, r., etc.).
5. **Retroflex** is very common sound in many Pakistani languages which is made by curling the tip of the tongue up and back so that the tongue tip moves during the retroflex sounds such as [ɳ, ɳ̌, ɳ̍, ɳ̎, ɳ̏].
6. **Palato-alveolar** and palatal are also possible articulatory gestures commonly found in world languages. Similarly, velar sounds found in Urdu and other Pakistani languages need to be mentioned here including [x, ɣ] which are velar fricatives. The gestures for pharyngeal (such as Arabic pharyngeal fricative [ʕ] and epiglottal sounds (such as epiglottal fricative [ʕ̥] involve pulling the root of the tongue or the epiglottis back toward the back wall of the pharynx.

### Q. Plotting vowel on chart

Take the measurement of the first two formants and plot those values on a chart using the Excel spreadsheet. By putting F1 and F2 in separate columns, write the formant values associated with different vowels (giving vowels in the first column, the difference between F2 and F1 in the second column and F1 in the third). After putting the data in Excel sheet, use the Scatter chart from the same spreadsheet. Further in order to make it corresponding with the required values for F1 and F2, reverse the values for both formants (on both axis – Y and X). Now the zero for both F1 and F2 is at the right corner. Watch the video and you will find how F1 is inversely related to the height of the vowel and the difference between F2 and F1 to the frontness of the vowels. Once completed, export the chart to your Word document and give it the number and title accordingly.

### Q. Role of memory for speech=2 marks

Speech is quite diverse and complex particularly when it comes to the phonetics of individual. It is understandable that different speakers of the same language will have somewhat different productions of speech depending upon their vocal tract physiology and their own habits of speech motor coordination and more importantly due to their memory of speech. As per the phonetic implementation view, words are stored in speech memory in their most basic phonetic form and used when needed.

### Q. Tonal language:

Tone (in phonetics and phonology) as a suprasegmental feature refers to an identifiable movement (variation) or level of pitch that is used in a linguistically contrastive way. In tone (tonal) languages, the linguistic function of tone is to change the meaning of a word. For example, in Mandarin Chinese, [ma] said with a high pitch means ‘mother’ while [ma] said on a low rising tone means ‘hemp’. In other (non-tonal) languages, tone forms the central part of intonation, and the difference between, for example, a rising and a falling tone on a particular word may cause a different interpretation of the sentence in which it occurs. In the case of tone languages, it is usual to identify tones as being a property of individual syllables, whereas an intonational tone may be spread over many syllables. In the analysis of English intonation, tone refers to one of the pitch possibilities for the tonic (or nuclear) syllable. For further analysis, a set of four types of tone is usually used (fall, rise, fall–rise and rise–fall) though others are also suggested by various experts.

### Q. Stress time language:

It is sometimes claimed by the experts that different languages (and dialects) have different types of rhythmic patterns. Languages of the world are; therefore, divided into two broad categories: stress timed language and syllable timed languages. But this is basically the division of languages on the basis of their modes of timing (i.e., stress vs. syllable timed languages). Stress timed languages have stress as their dominating rhythmic feature meaning that these languages seem to be timed according to the stressed patterns (the division among the syllables is made on the basis of stressed and unstressed patterns e.g., English and German languages). In other words, in stress

timed languages, stressed syllables occur with regular intervals and their units of timing are perceived accordingly. Stress-timed rhythm is one of these rhythmical types, and is said to be characterized by a tendency for stressed syllables to occur at equal intervals of time.

### **Q. Nasalization:**

The speakers of Urdu, Punjabi and many other Pakistani regional languages learn to produce a variety of nasal vowels as the part of their mother tongue and face no issue in learning nasalization in vowels. However, for the speakers of other languages (such as English which does not have nasal vowels) have to learn this feature of vowels by starting saying the low vowel /æ/ as in man and by keeping the soft palate lowered. any languages have contrasts between nasal and oral vowels including French and Urdu. Urdu and Punjabi have many nasal vowels. Urdu has seven nasal vowels such as /he/ (meaning 'is' vs. /h / (meaning 'are' . Nasalization in vowels is a common feature of Indo Aryan languages. In IPA chart, the diacritic used for nasalization is the symbol [̃] called tilde (used above the phonetic symbol to show the nasality).

### **Q. Difference Between acoustic and auditory phonetics ....3**

Acoustic Phonetics is the study of detailed physical properties of sound we produce. It generally uses tools which read the changes in air pressure that the sound creates. Each sound has its different sound quality, which depends on the source filter, that is our speech organs. Each sound has its own f0, f1, f2 and f3 (formants), depending on the source modifier. f0 deals with the fundamental frequency, f1 gives the information about the pharyngeal cavity, f2 about the oral cavity and f3 about the position of the lips while the sound was produced. Auditory phonetics is just the other side of the coin for this study. It deals with the study of these articulated sound characteristics from the perception perspective. Auditory phonetics deals with the listener at a broader aspect. The perceived f0 is measured in terms of pitch and calculated in Mel or bark scales.

### **Q. Secondary cardinal vowels:**

Explaining vowels (particularly, the vowels of Non-European languages), a set of secondary cardinal vowels (as a precise set of references) was introduced by the same British phonetician Daniel Jones (1881-1967). Secondary cardinal vowels are easy to understand in connection with the primary cardinal vowel system. The main difference between primary and secondary cardinal vowels is related to lip-rounding as in some languages the feature of lip-rounding is possible for front vowels. By reversing the lip position (in comparison with primary cardinal vowels), the secondary series of vowel types is produced (e.g., rounding the lips for the front vowels).